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(1) Research Program/Project/Study Title: Library Attendance Monitoring System	
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Program or Project has component (if no, proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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☐ Applied ☐ Tech. Promotion/Commercialization	☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(9A) Abstract

With a library attendance monitoring system, number of visitors can be automatically recorded, most popular library services can be easily identified, and queues can be monitored and reduced. Likewise, paths people take through the library can be tracked, the busiest days and times can be determined, and reports can be generated easily. The researchers developed and designed a software called "Library Attendance Monitoring System" that will be used to record attendance of library users, track in real-time library space utilization and generate required reports. The study was conducted at Capiz State University, Pontevedra Capiz from June 2023 to September 2023. The development of the Library Attendance Monitoring System made use of the Systems Development Life Cycle (SDLC)-Iterative Development Model as guide for the entire duration of project development. It also utilized MS Visual Studio, C# Programming Language, and SQL Server for its database. The system features consisted the following: system security, updating of records, ranking module, and report generation. The study has undergone acceptability testing with the students of the campus and librarians of other campuses as the respondents. The system was highly accepted in terms of its product quality and quality in use which implies that the campus library and other libraries may utilize the software to enhance its delivery of services to its clientele.

(9B) Keywords

automated attendance; library software; library statistics; systems development life cycle; iterative development model

(10) Introduction

Statistical information is a vital tool for the management and development of organizations. In library setting, keeping statistics of activities is basic to the survival and progress of library services, in the measurement of its performance, and in identifying areas needing improvement. Library uses statistics for the purpose of preparing reports, assessing the functionality of different library services, determining budget allocation and collection usage, and in evaluating short and long-term plans.

The CAPSU Library management exerts so much effort on the library statistics that it compiles as the basis of its research and development in support of its responsibilities and functions. Library personnel spend most of their time in tallying and manually counting the number of users. Also, students and other library users

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take time in writing on the library attendance sheets when they enter or utilize library services.

Nowadays, more libraries are utilizing library monitoring system to improve services. With a library automated monitoring system, number of visitors can be automatically recorded, most popular library services can be easily identified, and queues can be monitored and reduced. Likewise, paths people take through the library can be tracked, the busiest days and times can be determined, and reports can be generated easily.

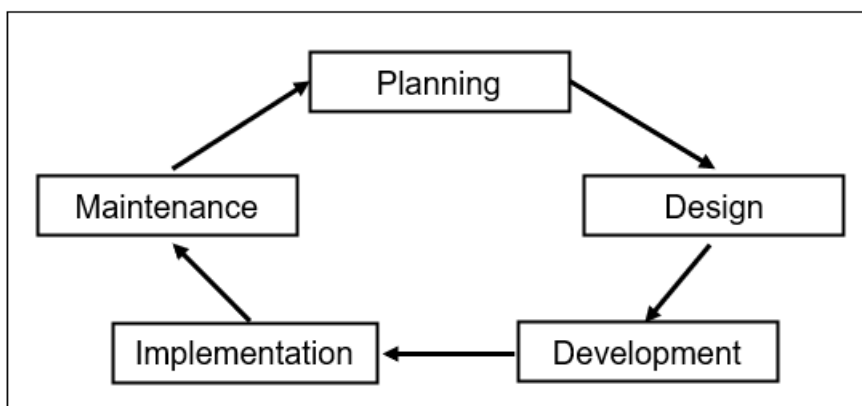
The library managers and library users enjoy the ease and convenience that these systems offer, but these are very expensive. Also, these systems are not customized to suit the needs of our current library setting.

This project is a monitoring system that is used to record attendance of library users, track in real-time library space utilization and generate required reports. The project was developed using C# programming language and MS SQL for database, and the use of barcode scanner as login device. The barcode reader is placed at the library entrance and connected to the computer system where this software is running. Library users have to simply scan library card at the barcode scanner to register their attendance at the time of entry and exit.

(11) Methodology

The development of the Library Attendance Monitoring System made use of the Systems Development Life Cycle (SDLC) Model.

The Systems Development Life Cycle is a multi-step, iterative process, structured in a methodical way. This process is used to model or provide a framework for technical and non-technical activities to deliver a quality system, which meets or exceeds a business' expectation or manage decision-making progression. The SDLC is broken down into the following phases: Systems Planning, Systems Analysis, Design Phase, System Implementation and Systems Operation, Support and Security.



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Figure 1. Systems Development Life Cycle (SDLC)

Different versions of SDLC continue to emerge in the IT world today. Among these are hybrid forms of SDLC, which have been created to suit the needs of different organizations. For this project, the proponents chose the Iterative Development Model – a subset of SDLC as guide for the entire duration of project development.

Iterative Development Model

In iterative model, iterative process starts with a simple implementation of a small set of the software requirements and iteratively enhances the evolving versions until the complete system is implemented and ready to be deployed. At each iteration, design modifications are made and new functional capabilities are added. The basic idea behind this method is also to develop a system through repeated cycles and in smaller portions at a time.

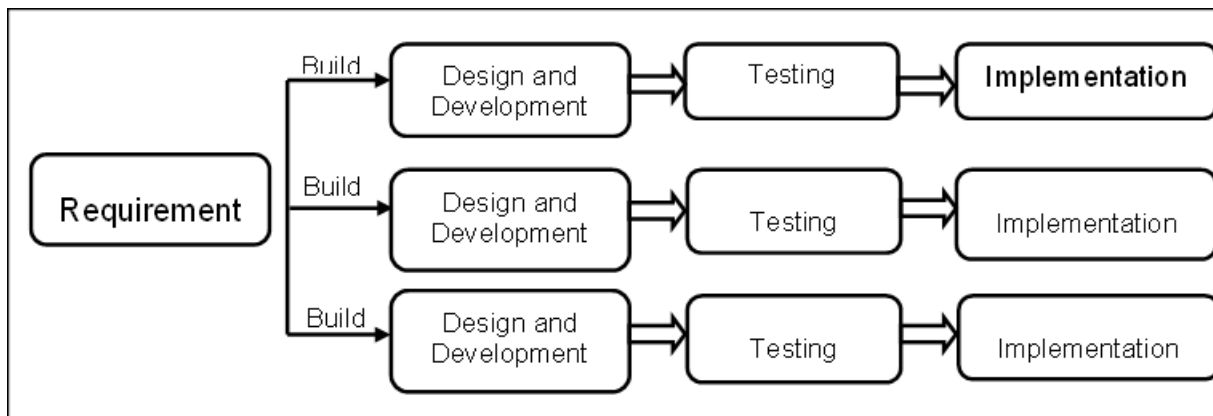


Figure 2. Iterative Development Model

(12) Results and Discussions

The researchers designed and developed a library attendance monitoring system which utilized MS Visual Studio, C# Programming Language, and SQL Server for its database.

System Overview

The Developed System (Library Attendance Monitoring System)

The researchers designed and developed a Library Attendance Monitoring System that allows the library personnel to simplify their work and easily and efficiently generate required reports, and provides convenience to the library users in logging-in and out upon entry and exit of the library.

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System Features

System Security

Log-in form was provided to meet the security of the system. To use this, the user is required to enter username and password before the system will open. The security interface was made through the use of MS Visual Studio using C# programming Language and SQL LocalDB. It has a user level like administrator and student. In the administrator level, the user can manage the system and can perform tasks like edit, update, delete and save records, and view reports while students can only perform login and logout.

Updating of Records

Correction and changes made on student records are easily done through the use of updating forms that allow changes on the data. The information stored in the database can be updated and manipulated using the forms provided.

Ranking Module

This feature allows the system to automatically generate the ranking of library users based on their attendance.

Report Generation

The Report Generation feature is found on the report form of the system wherein all the information that needs an output are automatically processed, displaying the print preview lay-out of the information. A record on library attendance based on the range of date specified according to course, time, and library service/section utilized can be automatically displayed.

Acceptability of the Library Attendance Monitoring System

Product Quality

Data in Table 1 reflect the acceptability level of the Library Attendance Monitoring System in terms of product quality which consists of Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. A grand mean of 4.58 was obtained which indicates that the respondents “highly accepted” the quality of the product.

Table 1. Product Quality

Product Quality	Mean	Verbal Interpretation
1. Suitability	4.52	Highly Accepted
2. Performance Efficiency	4.53	Highly Accepted

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3. Compatibility	4.58	Highly Accepted
4. Usability	4.63	Highly Accepted
5. Reliability	4.57	Highly Accepted
6. Security	4.58	Highly Accepted
7. Maintainability	4.62	Highly Accepted
8. Portability	4.60	Highly Accepted
GRAND MEAN	4.58	Highly Accepted

Note: Interpretation is based on the scale: 1.0-1.49 (Highly Unaccepted), 1.50-2.49 (Unaccepted), 2.50-3.49 (Moderately Accepted), 3.50-4.49 (Accepted), and 4.50-5.0 (Highly Accepted).

Quality in Use

Table 2 shows the acceptability level of Library Attendance Monitoring System in terms of quality in use consisting of Effectiveness, Efficiency, Satisfaction, Freedom from Risk, and Context Coverage. Data reveals that the respondents highly accepted the system in terms of its quality in use as shown by the computed grand mean of 4.59.

Table 2. Quality in Use

Quality in Use	Mean	Verbal Interpretation
1. Effectiveness	4.65	Highly Accepted
2. Efficiency	4.62	Highly Accepted
3. Satisfaction	4.64	Highly Accepted
4. Freedom from Risk	4.56	Highly Accepted
5. Context Coverage	4.50	Highly Accepted
GRAND MEAN	4.59	Highly Accepted

Note: Interpretation is based on the scale: 1.0-1.49 (Highly Unaccepted), 1.50-2.49 (Unaccepted), 2.50-3.49 (Moderately Accepted), 3.50-4.49 (Accepted), and 4.50-5.0 (Highly Accepted).

(13) Conclusions and Recommendations

Conclusions

The following conclusions were drawn:

1. The system has a login and logout feature that will monitor library users' attendance.
2. The system has designed a query module that will manage the searching of users' log and display real-time library space utilization.
3. The system has a feature intended for the generation of report on library users' attendance and ranking of top ten library users.

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4. In terms of the system's product quality and quality in use aspects, it was highly accepted by the respondents.

Recommendations

Based on the conclusions formulated, the researchers have the following recommendations:

1. The system may be used in the library for the efficient generation of reports and improvement of the delivery of services, and also for the convenience of library users for they will not be queueing and writing anymore in the log sheets.
2. A further study of the system may be made by adding more features not considered in the current software to expand its scope and make it more beneficial.
3. The utilization of the software may be replicated to other CAPSU campus/satellite college libraries or other school/academic libraries.

(14) Output

The output of this research is a software called Library Attendance Monitoring System (LAMS).

(15) Technology Transfer Activity Conducted

The Library Attendance Monitoring System (LAMS) will be made available to other campus/satellite and other school/academic libraries who wish to avail of the project.

(16) Economic Analysis

The project cost was minimal since the researchers only used printing supplies for the reproduction of the questionnaires. The barcode reader and the computer unit used was already available in the library. The library has already started reaping the benefits of the project since its utilization in September 2023.

(17) Acknowledgments

The researchers wish to convey their appreciation to the administration for the support, and to the students and librarians of other campuses who served as respondents of the acceptability testing of the system.

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